Exercise 1

We need to determine $T_k(n)$ for

$$T_k(n) = (k \cdot (k+2) - 3) \cdot T_k(n/5) + \Theta\left(n^{1+\frac{k}{2}-\frac{1}{2}\cdot(k \bmod 4)}\right)$$

We apply the Master Theorem which distinguishes between the values of $\log_5(k(k+2)-3)$ for $k = 2, 3, 5, 8$. We need to check the conditions for the application of the Master Theorem in each of these cases

$k=2$: $\gamma := \log_5 5 = 1$

$f(n) = \Theta(n)$

We apply case II of MT

$T_2(n) = \Theta(n \log n)$ and then

$g_2(n) = n \log n$

$k=3$: $\gamma := \log_5 12$

$f(n) = \Theta(n)$

We apply case I of MT

$T_3(n) = \Theta(n^{\log_5 12})$ and then

$g_3(n) = n^{\log_5 12}$

$k=5$: $\gamma := \log_5 32$

$f(n) = \Theta(n^3)$

We apply case III of MT

$T_5(n) = \Theta(n^3)$

Regularity condition

$g_5(n) = n^3$

$$32 f\left(\frac{n}{5}\right) \leq C f(n)$$

$$32 \cdot \left(\frac{n}{5}\right)^3 \leq C \cdot n^3$$

$$\frac{32}{125} n^3 \leq C \cdot n^3 \qquad C = \frac{32}{125}$$

$k=8$: $\gamma := \log_5 77$

$f(n) = \Theta(n^5)$

We apply case III of MT

$T_8(n) = \Theta(n^5)$

$g_8(n) = n^5$

Regularity condition

$$77 f\left(\frac{n}{5}\right) \leq C \cdot f(n)$$

$$32 \cdot \left(\frac{n}{5}\right)^5 \leq C \cdot n^5$$

$$\frac{32}{3125} n^5 \leq C n^5 \qquad C = \frac{32}{3125}$$

Exercise 2.

a) We apply the Master Theorem with

$r := \log_2 3$

$f(n) = \Theta(n^2)$

We apply case $\underline{III}$ of the MT

$$3\left(\frac{n}{2}\right)^2\left(\cos\frac{n}{2}+1+\varepsilon\right) \leq C\cdot n^2(\cos n+1+\varepsilon)$$

We know that for arbitrary n

$$\cos\frac{n}{2} \leq 1 \qquad \cos n \geq -1$$

$$3\left(\frac{n}{2}\right)^2(1+1+\varepsilon) \leq C\cdot n^2(-1+1+\varepsilon)$$

$$3\cdot\frac{n^2}{4}(2+\varepsilon) \leq C\cdot n^2\cdot\varepsilon$$

and let's say $\varepsilon \leq 1$

$$3\cdot\frac{n^2}{4}\cdot 3 \leq Cn^2$$

then $C \geq \frac{9}{4}$ but we should have $C < 1$ so the theorem cannot be applied.

b) We apply the MT with $\gamma := \log_2 3$
$f(n) = \mathcal{O}(n^2)$

We apply case III of the MT and check the condition

$$3 \cdot \left(\frac{n}{2}\right)^2 \leq C \cdot n^2$$

if $C \geq \frac{3}{4}$ the condition is satisfied.

So we conclude $S(n) = \Theta(n^2)$, so
$g(n) = n^2$

c) We prove $T(n) = \mathcal{O}(g(n)) \wedge T(n) = \Omega(g(n))$

1. $T(n) = \mathcal{O}(g(n))$

We prove $T(n) \leq cn^2$ for some $c > 0$
for $n > 3$:

$$\begin{aligned} T(n) &= 3T\left(\frac{n}{2}\right) + n^2(\cos n + 1 + \varepsilon) \\ &\overset{IH}{\leq} 3c\left(\frac{n}{2}\right)^2 + n^2(\cos n + 1 + \varepsilon) \\ &\overset{*}{\leq} \frac{3}{4}cn^2 + 3n^2 \\ &= n^2\left(\frac{3c}{4} + 3\right) \\ &\leq cn^2 \end{aligned}$$

(*) $\frac{3c}{4} + 3 \leq c$, so for $c \geq 12$

2. $T(n) = \Omega(g(n))$

We prove $T(n) \geq cn^2$ for some $c > 0$ for $n > 3$.

$$T(n) = 3T\left(\frac{n}{2}\right) + n^2(\cos n + 1 + \varepsilon)$$
$$\overset{IH}{\geq} 3c\left(\frac{n}{2}\right)^2 + n^2(\cos n + 1 + \varepsilon)$$
$$\geq \frac{3}{4}cn^2 + n^2$$
$$= n^2\left(\frac{3}{4}c + 1\right)$$
$$\overset{(*)}{\geq} cn^2$$

(*) $\frac{3}{4}c + 1 \geq c$, so for $c \leq 4$

Exercise 3

In order to show that we can't apply Masters Theorem to the recurrence, we need to show that for each of the cases the conditions imposed over $f(n) = n^3(n \bmod 2)$ can't be satisfied. If at least one of the conditions ~~doesn't~~ holds then master theorem can ~~cannot~~ be applied.

$\gamma := \log_2 8 = 3$

Case I

Suppose $f(n) = O(n^{3-\varepsilon})$ for some $\varepsilon > 0$

$n \bmod k \in \{0, 1, 2, \dots, k-1\}$

for $n \to \infty$ $f(n) \approx (k-1)n^3$ which is not slower than n^3, therefor can't be $O(n^{3-\varepsilon})$

Case II

Suppose $f(n) = \Theta(n^3) \approx cn^3$
For infinitely many n, the function is zero, therefore it cannot be bounded by $\Omega(n^3)$.

Case III

Suppose $f(n) = \Omega(n^{3+\varepsilon})$ for some $\varepsilon > 0$

For infinitely many n, the function is zero, therefore we can't say that $f(n) \geq cn^{3+\varepsilon}$

None of the conditions hold therefore MT cannot be applied.